Abhilash Reddy Somigari

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EDUCATION

West Virginia University, M.S. in Computer Science – GPA 3.87/4.00 Jan 2025 – Dec 2026 (Expected)
Coursework: Agentic AI, Deep Learning, Pattern Recognition, Big Data

Jawaharlal Nehru Technological University Hyderabad, B.Tech in Computer Science (CSE) Nov 2020 – Jun 2024

Certifications: Oracle Agentic AI Certified Foundations Associate, IBM RAG and Agentic AI Professional Certificate

SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL
AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLOv5, SAM
Generative AI: LLMs, RAG, AI Agents, LangGraph, LangChain, MCP, OpenAI & Anthropic APIs, LLM Evaluation
Backend & Databases: FastAPI, Flask, Django, REST APIs, React, Next.js, PostgreSQL, pgvector, Chroma, Pandas
Cloud & DevOps: AWS (EKS, ECR, RDS), Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Linux

EXPERIENCE

Graduate Research Assistant, West Virginia University – Morgantown, WV Aug 2025 – Aug 2026

- Cut manual heatmap inspection by **70%** with a **custom XAI faithfulness metric** (localization-curve AUC) that ranks Grad-CAM, EigenCAM, and Integrated Gradients on a YOLOv5 detector trained on hand-labeled satellite imagery
- Cut experiment iteration time by **50%** with a **full-stack web app** (Flask, React) that runs detection and XAI in-browser
- Built a **SAM3 video segmentation pipeline** over **20** laparoscopic surgeries; scaling-based normalization made cross-frame comparisons **30%** more reliable, and automated cut-ratio analysis cut manual review **40%**

Teaching Assistant (Data-Driven WV), West Virginia University – Morgantown, WV Jan 2026 – May 2026

- Cut student setup time by **30%** and raised submission success by **35%** via GOES-16 data pipelines and starter code
- Designed spatiotemporal ML assignments (**CNN and ConvLSTM** cloud forecasting vs. a persistence baseline in PyTorch), improving evaluation consistency by **25%**

Software Developer Intern, SAK Informatics – Hyderabad, Telangana Jan 2024 – Mar 2024

- Reduced manual data handling by **30%** by building Python automation scripts and data pipelines in Agile sprints
- Built **Flask REST APIs** and internal tools (Pandas, NumPy), making data access and prototyping **25%** faster

Mentor, Smart Interviews – Hyderabad, Telangana Oct 2023 – Dec 2023

- Mentored **80+ students** in data structures and algorithms through lectures, live coding, and doubt-solving sessions

PROJECTS

Rootline: AI Incident-Response Agent | *Python, LangGraph, FastAPI, PostgreSQL, OpenTelemetry* rootline

- Built an **AI SRE agent in LangGraph** that root-causes incidents with 5 evidence tools (metrics, logs, traces, deploys, RAG), reaching **93% accuracy vs. a 30% LLM-only baseline** on a 30-incident, 3-microservice benchmark (3 runs, $0.03 each)
- Found and fixed **4 flaws in the eval harness** (leaked fault telemetry, cross-case contamination, keyword scoring) and used audit trails to fix an agent that **stopped at the first plausible cause**
- Secured the **human-approval gate** with role-based tokens and an **atomic Postgres compare-and-set** (1 of 20 concurrent approvals executes): **0 unsafe actions in 90 investigations**, 47 pytest tests in CI

EvalForge: CI/CD Evaluation Harness for AI Agents | *Python, FastAPI, PostgreSQL, PyPI* evalforge

- Published **evalforge-agents** to PyPI: concurrent YAML scenario suites for tool-using agents with **11 graders** (tool sequence, citation groundedness, LLM-as-judge, human scores) and **104 unit and Postgres integration tests**
- Benchmarked **GPT-4o vs. Claude Opus 5.5** (24 trials each), finding a **4.7x cost / 3.6x latency** trade-off and a refund-handling gap that aggregate scores missed; CI baseline gates block regressions per branch
- Closed an **unthrottled API-key-guessing path** with Postgres-backed rate limits shared across replicas; releases ship via hash-pinned builds and **OIDC trusted publishing**

Citewise: Self-Improving SDR Agent | *Python, LangGraph, FastAPI, pgvector, Terraform* citewise

- Architected **multi-agent orchestration in LangGraph** (prospect, enrich, verify, write/critique, send) on FastAPI, Postgres/pgvector, OpenTelemetry, and Terraform; verified **8 of 9** leads at confidence ≥ 0.7 without SMTP probing
- Engineered an **EvalForge-gated self-improvement loop** (failure clustering, learnings memory, prompt optimizer, same-seed A/B, regression gates) and ran **960 seeded eval trials** (3 rounds $\times$ 4 variants $\times$ 80) on a local 7B model
- Added **paired bootstrap confidence intervals** showing the gains (+3, +2, +1 of 80) were noise; shipped **significance-gated promotion** that would have rejected all three and published the null result